

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**

**Department of Computer Science and Engineering**

**CO2 - ASSESSMENT TOOL 2 – Architecture Design Assignment**

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|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------|
| <b>Course Code:</b>   | CSA1007                                                                                                                                         | <b>Course Title:</b> | Software Engineering                                     |
| <b>CO Assessed:</b>   | CO2                                                                                                                                             | <b>Assessment:</b>   | Assessment Tool 2 –<br>Architecture Design<br>Assignment |
| <b>Weightage:</b>     | 20%                                                                                                                                             | <b>Total Marks:</b>  | 25                                                       |
| <b>CO2 Statement:</b> | <i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i> |                      |                                                          |
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| <b>Date:</b>          | 30/07/2026                                                                                                                                      |                      |                                                          |

**Code of Conduct**

*I \_\_\_\_\_P.Sonu(192365057)\_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: \_\_\_\_\_P.Sonu\_\_\_\_\_

## 1. Introduction

The **Smart Public Library Management Platform** is an intelligent distributed software system designed to automate and modernize public library operations across multiple branches. Traditional library systems rely on manual record keeping or isolated software applications, making it difficult to manage thousands of books, digital resources, memberships, and borrowing activities efficiently. These limitations often lead to misplaced books, delayed returns, inaccurate inventory records, and poor resource utilization.

The proposed platform provides a centralized cloud-based solution that connects multiple library branches in real time. It enables members to search for books, reserve unavailable titles, borrow digital resources, receive due-date reminders, and access personalized book recommendations. Librarians can efficiently manage inventory, memberships, circulation activities, and digital resources through an integrated dashboard, while administrators gain access to detailed analytics and utilization reports for effective decision-making.

The platform also incorporates intelligent features such as machine learning-based book recommendations, demand prediction, and automated notifications, making the library system more efficient, scalable, and user-friendly.

## 2. System Requirement Analysis

System requirement analysis is the process of understanding the business needs, identifying system functionalities, and defining quality requirements that influence the software architecture.

The Smart Public Library Management Platform is designed to solve the operational problems of traditional libraries while supporting future expansion. Since the platform connects multiple library branches, the architecture must support distributed operations, cloud storage, high availability, and secure communication between different modules.

The system must efficiently manage:

- Library memberships
- Book catalog
- Digital resources
- Book issue and return
- Online reservations
- Inventory tracking
- Notifications
- Reports
- Recommendation engine
- Demand prediction

The platform should support thousands of users simultaneously while maintaining fast response time and secure access.

## 2.1 Existing Challenges

The current library management process faces several problems:

- Manual updating of book records.
- Difficulty locating available books.
- No centralized inventory management.
- Limited support for digital resources.
- Delayed notifications for due dates.
- No online reservation facility.
- No recommendation system.
- Difficulty managing multiple library branches.
- Poor reporting and analytics.
- High operational workload.

These challenges directly influence the selection of software architecture. The Smart Public Library Management Platform is an intelligent distributed software system designed to automate and modernize public library operations across multiple branches. Traditional library systems rely on manual record keeping or isolated software applications, making it difficult to manage thousands of books, digital resources, memberships, and borrowing activities efficiently. These limitations often lead to misplaced books, delayed returns, inaccurate inventory records, and poor resource utilization.

## 3. System Objectives

The major objectives of the Smart Public Library Management Platform are:

- Automate library operations.
- Manage books across multiple branches.
- Provide centralized inventory tracking.
- Enable online reservations.
- Improve book circulation.
- Reduce overdue returns.
- Recommend books using Artificial Intelligence.
- Predict future book demand.
- Generate analytical reports.
- Improve overall user experience.
- Increase operational efficiency.
- Support cloud-based deployment.

## 4. Functional Requirements

The architecture should support the following functional requirements.

### FR1 User Registration

Members should be able to create new library accounts.

**FR2 Authentication**

Secure login for Members, Librarians, and Administrators.

**FR3 Membership Management**

Approve, renew, suspend, and manage memberships.

**FR4 Book Management**

Add, update, delete, and organize books.

**FR5 Book Search**

Search books using multiple filters.

**FR6 Book Issue**

Issue books to registered members.

**FR7 Book Return**

Return books and update inventory.

**FR8 Online Reservation**

Reserve unavailable books.

**FR9 Digital Library**

Provide access to e-books and journals.

**FR10 Notifications**

Send reminders through Email and SMS.

**FR11 Recommendation System**

Recommend books according to user interests.

**FR12 Demand Prediction**

Predict future demand using historical data.

**FR13 Inventory Tracking**

Track books across all branches.

**FR14 Analytics**

Generate reports regarding:

- Book circulation
- Branch performance
- Popular books
- Member activities

**5. Non-Functional Requirements**

The architecture must satisfy the following quality requirements.

### **Performance**

- Search response below **2 seconds**
- Report generation below **10 seconds**
- Support **10,000 concurrent users**

## **Selection of Software Architecture**

Choosing the appropriate software architecture is one of the most important decisions during software development because it determines the system's scalability, maintainability, performance, security, and future growth. The Smart Public Library Management Platform is expected to support multiple library branches, thousands of users, cloud deployment, digital resources, AI-based recommendations, demand prediction, and real-time notifications. Therefore, the architecture must be flexible enough to accommodate future enhancements without affecting existing services.

After analyzing the system requirements, the most suitable architecture for this project is a Hybrid Architecture, which combines Layered Architecture and Microservices Architecture.

### **Why Hybrid Architecture?**

A Hybrid Architecture combines the advantages of both Layered and Microservices architectures. The Layered Architecture provides a clear separation of responsibilities between presentation, business logic, and data access layers, making the system easier to develop and maintain. At the same time, the Microservices Architecture divides the application into independent services that can be developed, deployed, and scaled separately.

This approach is highly suitable because different modules such as user management, book management, reservation management, notification services, recommendation systems, and analytics have different workloads. Independent deployment allows each module to scale according to demand without affecting the entire application.

## **Justification for Selecting Hybrid Architecture**

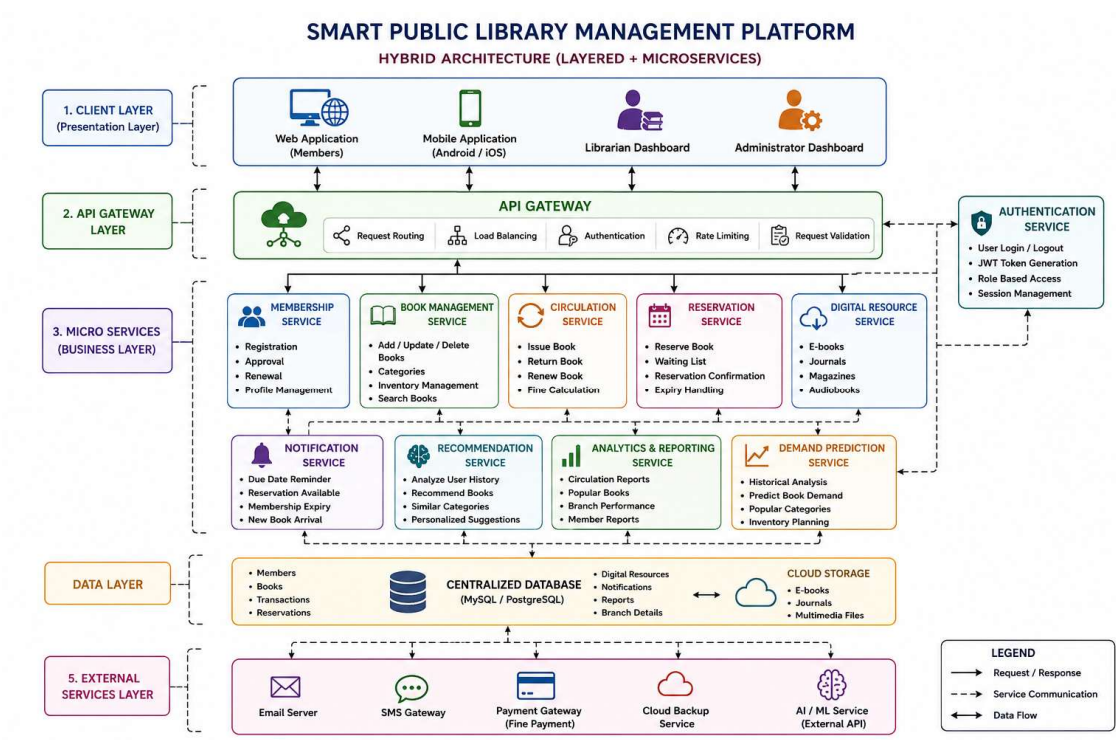
The Hybrid Architecture is selected for several technical and business reasons.

Firstly, the Smart Public Library Management Platform supports multiple branches located in different geographical locations. Microservices allow each functional module to communicate through APIs while remaining independent, ensuring smooth operations across all branches.

Secondly, some modules such as the Recommendation Engine and Demand Prediction use Artificial Intelligence and Machine Learning. These modules require higher computational resources compared to normal book management operations. Microservices allow these services to scale independently.

Thirdly, cloud deployment becomes easier because each service can be deployed separately using containers such as Docker and orchestration tools like Kubernetes.

Finally, the Layered Architecture improves code organization by separating the user interface, business logic, and database operations, making the system easier to maintain and extend.



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The platform also incorporates intelligent features such as machine learning-based book recommendations, demand prediction, and automated notifications, making the library system more efficient, scalable, and user-friendly.

System requirement analysis is the process of understanding the business needs, identifying system functionalities, and defining quality requirements that influence the software architecture.

The Smart Public Library Management Platform is designed to solve the operational problems of traditional libraries while supporting future expansion. Since the platform connects multiple library branches, the architecture must support distributed operations, cloud storage, high availability, and secure communication between different modules.

## Future Enhancements

The Smart Public Library Management Platform is designed with a scalable architecture that supports future expansion and integration of advanced technologies. As library services continue to evolve, several new features can be incorporated to improve user experience and operational efficiency.

One important enhancement is the implementation of RFID (Radio Frequency Identification) technology for automatic book tracking. RFID tags can help librarians quickly locate books, reduce manual effort, and improve inventory accuracy.

The system can also introduce QR code-based self-checkout and self-return kiosks, allowing members to borrow and return books independently without waiting for librarian assistance. This will reduce queues and improve service speed.

An AI-powered chatbot can be integrated to provide 24×7 assistance to library members. The chatbot can answer common questions, guide users in searching for books, explain library policies, and assist with reservations.

Future versions of the platform may include voice search and multilingual support, enabling users to search for books using voice commands and interact with the system in multiple languages. This feature will improve accessibility for a wider range of users.

The recommendation system can be enhanced using advanced Artificial Intelligence and Machine Learning algorithms to provide more accurate and personalized book suggestions based on reading habits, interests, and borrowing history.

A dedicated mobile application can be developed with additional features such as offline access to digital resources, push notifications, digital membership cards, and online fine payment through UPI, debit cards, or digital wallets.

The platform can also be integrated with national digital library networks, university libraries, and e-learning platforms, allowing users to access a larger collection of digital resources through a single application.

Cloud infrastructure can be further enhanced with auto-scaling, disaster recovery mechanisms, and distributed database replication to ensure uninterrupted service and improved system performance during high traffic.

These future enhancements will make the Smart Public Library Management Platform more intelligent, user-friendly, secure, and capable of meeting the growing needs of modern public libraries.

## **Conclusion**

The Smart Public Library Management Platform provides a modern, scalable, and intelligent solution for managing library operations across multiple branches. By adopting a Hybrid Architecture (Layered + Microservices), the system effectively supports core functionalities such as book management, membership management, reservations, circulation, digital resource access, notifications, analytics, and demand prediction while ensuring flexibility and maintainability.

The proposed architecture addresses key software quality attributes including scalability, security, performance, reliability, availability, and maintainability. Independent microservices allow different modules to be developed, deployed, and scaled separately, while the layered structure ensures clear separation of responsibilities and easier system maintenance. The use of cloud-based deployment, centralized databases, API Gateway, and secure authentication mechanisms enables reliable and efficient communication between all components.

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